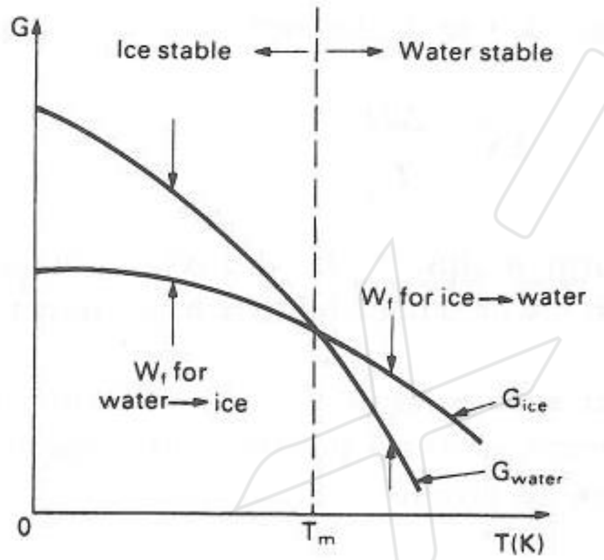
1. Basics: phase transformation

1. A touch of thermodynamics: driving force in phase transformation

System will tend towards thermodynamic equilibrium, at which the Gibbs energy G, is at a minimum. And ΔG is Gibbs free energy. When ΔG is negative, there is a driving force for system change.

For example, for pure water at 0°C (T_m), there is no change in Gibbs energy when the solid becomes a liquid. We have neutral equilibrium.
For pure water at $T < T_m$, $G^s < G^L$. There is a negative ΔG if the liquid becomes a solid, leading to an unstable situation and a driving force for this change.



$$\Delta G = \Delta H - T\Delta S$$

For coarsening:

$$V_{final} = V_{initial}$$

$$\Delta G = (A\gamma)_{final} - (A\gamma)_{initial}$$

ΔG can be also quantified. For example, coarsening is a phenomenon where small precipitates tend to merge with bigger precipitates as there is a large driving force for this transformation. ΔG can be calculated by the difference between total interfacial energy of the joined big particle and the total interfacial energy of the small particles.

2. Transformation rate influenced by temperature

ΔG can be expressed by the change of enthalpy (ΔH) and entropy (ΔS).
For a small departure from the melting point, the values of H and S can be considered independent of T. Therefore the driving force for transformation increase as the temperature decreases below the melting point.

$$\Delta G = \Delta H - T\Delta S \qquad \therefore 0 = \Delta H - T_m \Delta S$$

$$\therefore \Delta S = \frac{\Delta H}{T_m} \qquad \therefore \Delta G = \Delta H \left(\frac{T_m - T}{T_m}\right)$$

$$\Delta G = \Delta H - T\Delta S$$

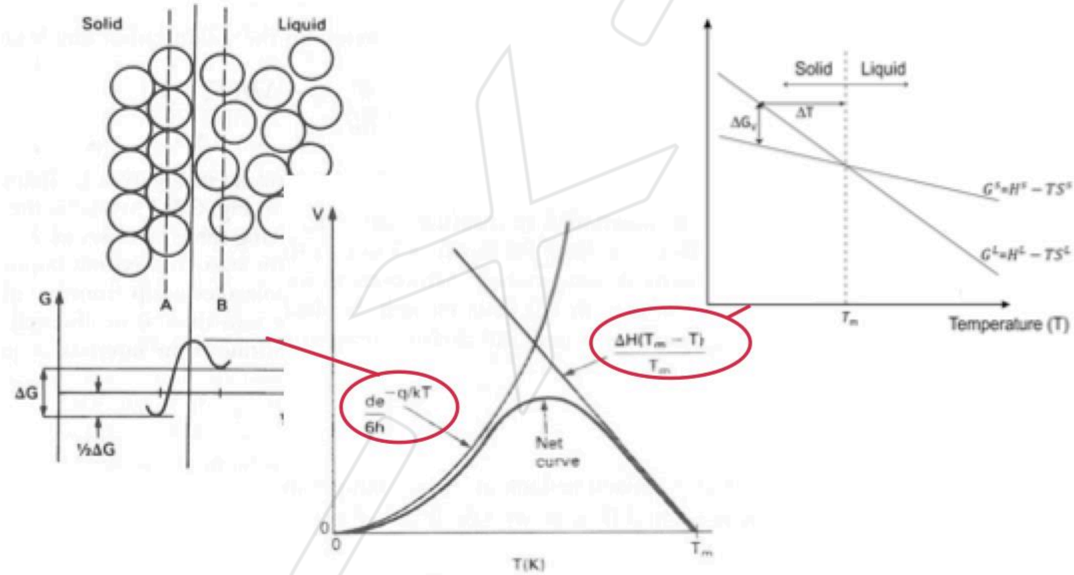
$$0 = \Delta H - T_m \Delta S \qquad \Delta G = \Delta H - T\frac{\Delta H}{T_m}$$

$$\Delta S = \frac{\Delta H}{T_m} \qquad \Delta G = \Delta H\left(\frac{T_m - T}{T_m}\right)$$

Undercooling

On the other hand, the probability that an atom will jump from the liquid to the solid ___decrease___ (the ___mobility___ of the atoms) as the temperature decreases.

Therefore as shown below the transformation rate ___increase___ as the temperature decreases due to the ___increase___ driving force. Then the rate decreases as the low temperature ___reduces___ atomic motion, giving an optimum temperature where transformation rate is maximal.

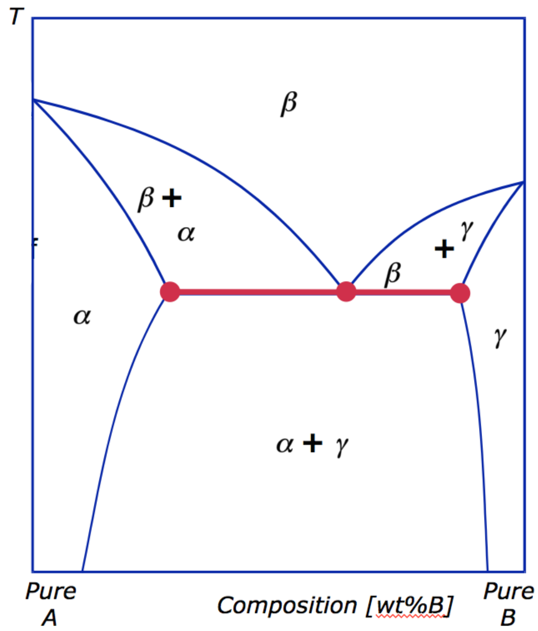


3. Diffusion controlled phase transformation

Diffusion is mass transport by atomic motion. In many cases, a phase transformation requires the diffusion of solute away from a growing phase, such as

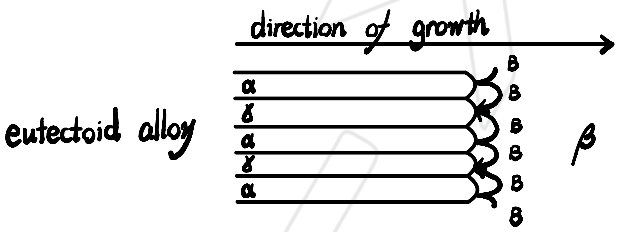
Eutectic growth: ___ $L \longrightarrow \alpha + \beta$ ___

Eutectoid growth: ___ $\beta \longrightarrow \alpha + \gamma$ ___



For example, for eutectoid growth, α is a phase containing a large amount of _A_ and small amount of _B_ ; while ϒ is a phase containing large amount of _B_ and small amount of _A_. In this case the speed of reaction is determined by the rate of transport of solute from that phase. The _laminar_ geometry of the phases is also formed because this is the fastest way for B to transform from phase α to phase ϒ.

Sketch what microstructure of the eutectoid phase is expected during the phase transformation (including the B atoms moving direction).



4. **Nucleation of a new phase**

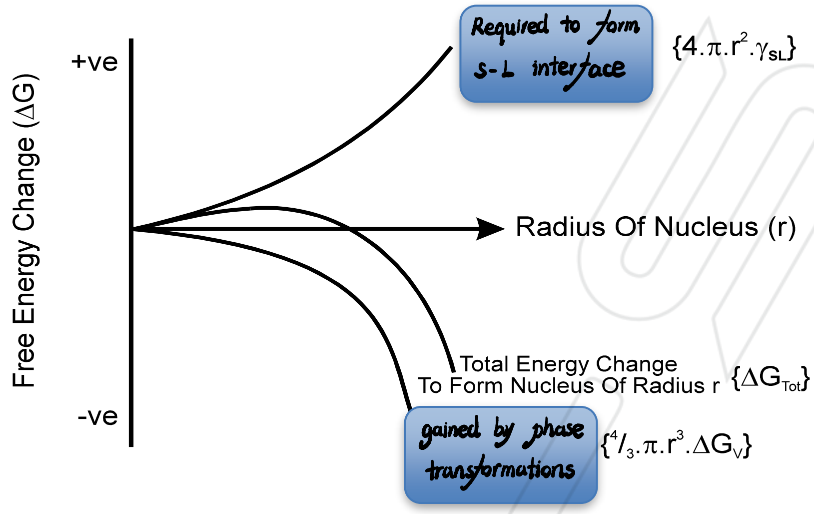
Homogenous nucleation theory

In this theory, the Gibbs free energy is influenced by the energy _required_ to form a new interface (_S-L interface_) and the energy _gained_ by phase transformation from liquid to solid.

Therefore near to the T_m, clusters of atoms with the solid structure (nuclei) will appear in the liquid but these nuclei will only grow if _increase_ their size will result in a lowering of Gibb`s free energy. So there is a _critical radius_ above which all nuclei will grow corresponds to the maxima in the total free energy.

Mark each part of the equation below in the diagram. What energy do they express.

$$\Delta G_{Tot} = \left(-\frac{4}{3}\pi r^3 \Delta G_V\right) + (4\pi r^2 \gamma_{SL})$$

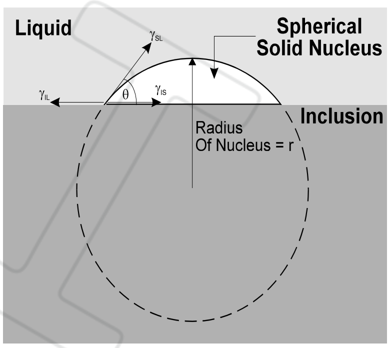


The critical radius is inversely proportional to the current level of __undercooling__ and it is extremely improbable that more atoms than needed for a 1nm sphere in the liquid will spontaneously form a solid arrangement. This translates into a required undercooling of $\Delta T \sim 100K$.

$$r_{crit} = \frac{2 \cdot \gamma_{SL}}{\Delta G_V}$$

Heterogeneous nucleation theory

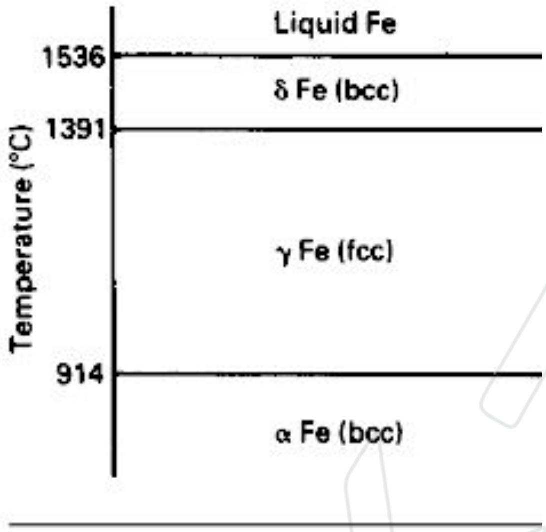
In reality nucleation occurs on the surface of __crucible / inclusion__. Here there already exists a S-L interface so when a nuclei forms the new S-L interface has __less__ interface, leading to __less__ required energy to create a new surface.



__Nucleants__ such as TiB_2 or Al_3Ti to help the heterogeneous nucleation.

Additionally, a solid phase contains many high energy ___defects___, e.g. ___dislocations___, grain boundaries. These regions of higher energy are ___good___ heterogeneous nucleation sites. Homogeneous nucleation is very rare in a solid phase.

5. Diffusive vs. displacive phase transformations in pure iron



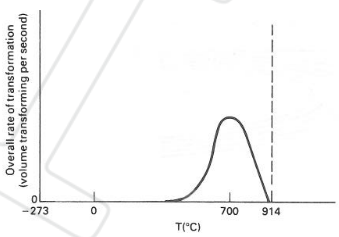
Iron (Fe) has ___polymorphism___. Ɣ-Fe and α-Fe are engineering related phases (more details see Topic Steel).

$\gamma \rightarrow \alpha$ diffusive transformation

Phase transformations are ___nucleation___ and ___growth processes___.
When we have a larger undercooling, the critical radius will ___smaller___, the driving force/ ___Gibbs free energy___ will be ___larger___, but the thermal energy for the nucleation will be ___smaller___, the mobility of the atom will be ___reduced___.

So the phase transformation rate is proportional to the numbers of α nuclei and the growth rate of α- Ɣ interface.

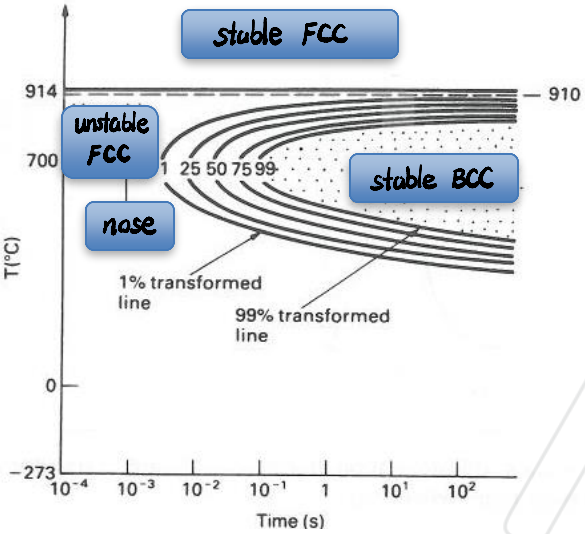
Transformation rate (volume s^{-1}) $\propto$ Number of α nuclei X Growth rate of α-γ interface



___Time - Temperature - Transformation___ (TTT) curves can be used to indicate at certain temperature and certain time, what the phase transformation status is.

Mark stable fcc iron, unstable fcc iron, stable bcc iron, "nose" in the curve. What does "25" on the curve mean?

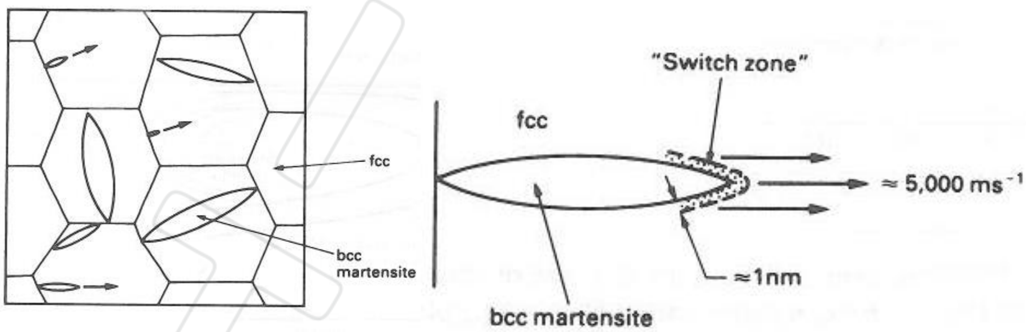
25% unstable FCC iron ⟶ stable BCC iron



$\gamma \rightarrow \alpha$ displacive transformation (Martensitic transformation)

In Fe, the system has another way to transform which is so called displacive transformation.

In this transformation, small lenses of BCC - Fe form at grain boundaries. These propagate across grains at the speed of sound. There is a "switch zone" in which atomic bonds are broken and remade to change from FCC to BCC and this transformation is similar to dislocation motion breaking and remaking bonds.



As lenses grow, they distort the lattice planes and some of the Gibbs free energy (ΔG) is absorbed as strain energy, therefore martensite-iron is very hard (more see the topic of Steel)

If all the available ΔG is absorbed as strain energy, the transformation stops. Therefore, there must be enough ΔG to start the displacive transformation (i.e. provide the strain energy). The volume fraction transformed then increased with decreasing temperature (increasing ΔG available to accommodate the strain energy).

In the Time - Temperature - Transformation curves below, process 1 gives a diffusive transformation and process 2 gives a displacive transformation. The image below presents the microstructure of BCC martensite growing. The arrow 1 points at FCC phase, and arrow 2 points at BCC martensite phase.